Characteristic Equation :

$A_{n\times n}$

$\det(A - xI_n)$ is said to be the characteristic polynomial of A.

$$\psi_A(x) = \det(A - xI_n)$$

$\boxed{\psi_A(x) = 0} \longrightarrow$ Characteristic equation

$$A = (a_{ij})_{n\times n}$$

$$\psi_A(x) = \det(A - xI_n)$$

$$= \begin{vmatrix} a_{11}-x & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22}-x & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn}-x \end{vmatrix}$$

$= c_0 x^n + c_1 x^{n-1} + \cdots + c_n$, where $c_0 = (-1)^n$

where, $c_0 = (-1)^n$

$c_r = (-1)^{n-r} \times \left(\underbrace{\text{Sum of the principal minors of } A \text{ of order } r}_{\text{Doubt}}\right)$

$$c_1 = (-1)^{n-1}(a_{11} + a_{22} + \cdots a_{nn})$$

$$= (-1)^{n-1} \text{ trace } A$$

$*$ $c_n = \det A$.

Ex) $\begin{pmatrix} 1 & -2 \\ 3 & 1 \end{pmatrix} \Rightarrow \left|\begin{pmatrix} 1-x & -2 \\ 3 & 1-x \end{pmatrix}\right| = (1-x)^2 + 6 = 0$

$= x^2 - 2x + 7 \Rightarrow A^2 - 2A + 7I = 0$

$$\psi_A(x) = \begin{vmatrix} a_{11}-x & a_{12} \\ a_{21} & a_{22}-x \end{vmatrix} = (a_{11}-x)(a_{22}-x) - a_{12}a_{21}$$

$$= (-1)^2 x^2 - \underbrace{(a_{11}+a_{22})}_{\text{Trace } A} x + \underbrace{a_{11}a_{22} - a_{12}a_{21}}_{\det A}$$

Cayley-Hamilton Theorem:

"Every square matrix satisfies its own characteristic equation"

* If A be an $n\times n$ matrix and the characteristic polynomials of A be

$$c_0x^n + c_1x^{n-1} + ---- + c_n \text{, then}$$

$$c_0A^n + c_1A^{n-1} + --- c_nI_n = 0$$

$$\cancel{A^2 + 2AA}$$

Ex) Use Caylay-Hamilton Theorem to compute A^{-1}, where

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix}$$

Sol^n: The characteristic equation of A is

$$\det(A - xI_2) = 0$$

or, $$\begin{vmatrix} 2-x & 1 \\ 3 & 5-x \end{vmatrix} = 0$$

or, $$x^2 - 7x + 7 = 0$$

By Cayley-Hamilton theorem,

$$A^2 - 7A + 7I_2 = 0$$

$$A^2 - 7A = -7I_2$$

$$A(A - 7I_2) = -7I_2$$

$$\frac{-1}{7}A(A - 7I_2) = I_2$$

This gives $$A^{-1} = \frac{-1}{7}(A - 7I_2)$$

$$= \frac{1}{7}\begin{pmatrix} 5 & -1 \\ -3 & 2 \end{pmatrix}$$

Eigen Value of $A_{n\times n}$:

Roots of the characteristic equation are called eigen value of A.

$$\begin{vmatrix} 1-x & 0 \\ 0 & 2-x \end{vmatrix} = (1-x)(2-x)$$

* If A be a singular matrix, 0 is an eigen value of A.
* The eigen value of a diagonal matrix ~~and~~ are its diagonal elements.

Theorem:

If λ be an eigen value of a non-singular matrix A, then λ^{-1} is an eigen value of A^{-1}.

Proof: Since A is non-singular, A^{-1} exists.

Since λ is an eigen value of A,

$$\det(A - \lambda I_n) = 0$$

$$\det(A^{-1} - \lambda^{-1} I_n) = \det A^{-1} \cdot \det(AA^{-1} - \lambda^{-1} A^{-1})$$
$$= (\det A)^{-1} \det(I_n - \lambda^{-1} A)$$
$$= (\det A)^{-1} (\lambda^{-1})^n \det(\lambda I_n - A)$$
$$= (\det A)^{-1} (\lambda^{-1})^n (-1)^n \underbrace{\det(A - \lambda I_n)}_{0}$$
$$= 0$$

Theorem:

If A and P be both $n \times n$ matrices and P be non-singular then A and $P^{-1}AP$ have the same eigen values.

Proof: $\det((P^{-1}AP) - x I_n)$

$$= \det(P^{-1}AP - x P^{-1}P)$$
$$= \det(P^{-1}(A - x I_n) P)$$
$$= (\det P^{-1}) \cdot \det(A - x I_n) \cdot (\det P)$$
$$= (\det P)^{-1} \cdot \det(A - x I_n) \cdot (\det P)$$
$$= \det(A - x I_n)$$

Ex 1) Let $A = \begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}$

Characteristic equation of A is

$$\det(A - x I_n) = \begin{vmatrix} 1-x & 3 \\ 4 & 5-x \end{vmatrix} = (1-x)(5-x) - 12$$

$$= x^2 - 6x - 7 = 0 \implies x = -1, 7$$

Ex-2) $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Sol^n: $x^2+1=0 \Rightarrow x = i, -i \longrightarrow$ Imaginary value.

Ex-3) $A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{pmatrix}$

Sol^n: $\begin{vmatrix} 1-x & -1 & 0 \\ 1 & 2-x & -1 \\ 3 & 2 & -2-x \end{vmatrix} = (1-x)(x^2-4+2) - 1(-3+x+2)$

$= x^2 - 2 - x^3 + 2x - x + 1$

$= -x^3 + x^2 + x - 1 = 0$

$x^2(-x+1) - 1(-x+1) = 0$

$(x^2-1)(1-x) = 0 \Rightarrow x = 1, -1$

$(x+1)(x-1)^2 = 0$

Eigen Vector of a Matrix:

Let A be $n \times n$ matrix over a field F. A non-null vector X belonging to $V_n(F)$ is said to be an eigen vector or characteristic vector of A if $\exists$ a scalar $\lambda \in F$ such that

$$AX = \lambda X \quad \text{holds}$$

or, $AX - \lambda X = 0$

$(A - \lambda I)X = 0$

$\Rightarrow \det(A - \lambda I_n) = 0$

$|A - \lambda I_n| = 0$

Ex) Let $A = \begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}$

Sol^n: $\det(A - \lambda I_2) = 0$

or $\begin{vmatrix} 1-\lambda & 3 \\ 4 & 5-\lambda \end{vmatrix} = 0$ or, $(1-\lambda)(5-\lambda) - 12 = 0$

$\lambda^2 - 6\lambda - 7 = 0$

$(\lambda - 7)(\lambda + 1) = 0$

$\lambda = 7, -1$

let $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ be an eigen vector corresponding to -1.

Then $AX = -X$ gives   Doubt

$$\left.\begin{aligned} 2x_1 + 3x_2 &= 0 \\ 4x_1 + 6x_2 &= 0 \end{aligned}\right\} \Rightarrow x_1 + \frac{3}{2}x_2 = 0$$

The solution of the system is $k\begin{pmatrix} -3/2 \\ 1 \end{pmatrix}$, where k is an arbitrary real no.

The eigen vectors are

$$k\begin{pmatrix} -3/2 \\ 1 \end{pmatrix} \cdot k \neq 0 \quad = \quad c\begin{pmatrix} -3 \\ 2 \end{pmatrix} \cdot \quad c \text{ is non-zero real}.$$

let $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ be an eigen vector corresponding to 7.

$$AX = 7X$$

$$\left.\begin{aligned} -6x_1 + 3x_2 &= 0 \\ 4x_1 - 2x_2 &= 0 \end{aligned}\right\} \quad x_2 = 2x_1 \Rightarrow \begin{pmatrix} c \\ 2c \end{pmatrix} = c\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad c \in R$$

Eigen Vectors are,

$$\cancel{c\begin{pmatrix} -3 \\ 2 \end{pmatrix}} : c\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad c \neq 0$$ Doubt

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Theorem: Two eigen vectors of a square matrix A over a field F, corresponding to two distinct eigen value of A are linearly independent.

~~Proof~~ Proof:

let $X_1 \rightarrow \lambda_1 \qquad AX_1 = \lambda_1 X_1$

$\quad\; X_2 \rightarrow \lambda_2 \qquad AX_2 = \lambda_2 X_2$

$$c_1X_1 + c_2X_2 = 0$$

$$c_1AX_1 + c_2AX_2 = 0$$

$$c_1\lambda_1X_1 + c_2\lambda_2X_2 = 0$$

$$c_1\lambda_1X_1 + \lambda_2(c_2X_2) = 0$$

$$c_1(\lambda_1 - \lambda_2)X_1 = 0$$

Theorem: The eigen vector of a $m \times n$ matrix $A_{m \times n}$ over a field F, corresponding to an eigen value $\lambda \in F$, together with the null vector form a vector space, a subspace of $V_n(F)$

Proof: Let S be the set of all eigen vectors of A corresponding to λ and let $X_1, X_2 \in S$.

Doubt

$$AX_1 = \lambda X_1, \quad AX_2 = \lambda X_2$$

$$A(X_1 + X_2) = \lambda X_1 + \lambda X_2$$
$$= \lambda (x_1 + x_2)$$

$$\Rightarrow X_1 + X_2 \in S$$

$$A(cX_1) = c(AX_1)$$
$$= c\lambda X_1$$
$$= \lambda(cX_1)$$

$$\Rightarrow cX_2 \in S$$

Definition: For an r-fold eigen value λ, r is called the algebraic multiplicity of λ and the rank of the characteristic subspace corresponding to λ is called the geometric multiplicity of λ

$1 \leq$ geometric multiplicity $\leq$ algebraic multiplicity.

Ex) Let $A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$ The characteristic equation of A is $\det(A - x I_3) = 0$

Or, $x^2(1-x) = 0$

$x = 0, 0, 1$

The eigen values are $0, 0, 1$

The eigen vectors corresponding to the eigen ~~vector~~ value 0 are non-null solutions of

$$x + y + z = 0$$
$$z = 0$$

The eigen vectors are $c\begin{pmatrix}1\\-1\\0\end{pmatrix}$, c is non-zero real no.

$AX = X$

or, $(A - I_3)X = 0$

$y + z = 0$

$x + 2y + z = 0$

$c(1,-1,1)$, $c \in \mathbb{R}$, $c \neq 0$.

---

1 is an eigen value of algebraic multiplicity 1.
Eigen vector corresponding to 1 is -
$c\begin{pmatrix}1\\-1\\1\end{pmatrix}$, $c \in \mathbb{R}$, $c \neq 0$. gm = 1.

~~1 ≤ AM ≤~~ $1 \leq GM \leq AM$

Ex) $A = \begin{pmatrix}1 & 1 & 1\\-1 & -1 & -1\\0 & 0 & 1\end{pmatrix}$

- Characteristic eqn is $x^2(1-x) = 0$
- Eigen values are $0, 0, 1$

0 is an eigen value of algebraic multiplicity 2

Eigen vector $c\begin{pmatrix}1\\-1\\0\end{pmatrix}$, $c \in \mathbb{R}$, $c \neq 0$

geometric multiplicity = 1

Theorem:

The eigen values of a real symmetric matrix are all real

Proof: $A_{n \times n}$

Let λ be an eigen value of A.

Then $\det(A - \lambda I_n) = 0$

Therefore $\exists$ non-null solutions of the homogenous system $(A - \lambda I_n)X = 0$

Let X_1 be one such solution :

Then $(A - \lambda I_n) X_1 = 0$

$$A X_1 = \lambda X_1$$

Taking transpose of the conjugate .

$$(\overline{AX_1})^t = (\overline{\lambda X_1})^t$$

or, $\bar{X_1}^t \cdot \bar{A_1}^t = \bar{\lambda} \cdot \bar{X_1}^t$

or, $\bar{X_1}^t \cdot A = \bar{\lambda} \cdot \bar{X_1}^t$

or, $\bar{X_1}^t A X_1 = \bar{\lambda} \bar{X_1}^t X_1$

or, $\bar{X_1}^t \lambda X_1 = \bar{\lambda} \bar{X_1}^t X_1$

or, $\lambda \bar{X_1}^t X_1 - \bar{\lambda} \bar{X_1}^t X_1 = 0$

or, $(\lambda - \bar{\lambda}) \bar{X_1}^t X_1 = 0$

or, $\lambda - \bar{\lambda} = 0$, $\Rightarrow \lambda = \bar{\lambda}$

$$(\overline{AX_1})^t = (\bar{A} \cdot \bar{X_1})^t$$

$$= \bar{X_1}^t . \bar{A}^t$$

$$\overline{z_1 z_2} = \bar{z_1} \bar{z_2}$$

<u>Theorem</u>: The eigen values of a real skew-symmetric matrix are purely imaginary or zero.

<u>Theorem</u>: The eigen vectors corresponding to two distinct eigen value of a real symmetric matrix are orthogonal

<u>Proof</u>: $A_{n \times n}$

$$\begin{array}{c|c} X_1 & X_2 \\ \lambda_1 & \lambda_2 \end{array}$$

$A X_1 = \lambda_1 X_1$ $\qquad$ $A X_2 = \lambda_2 X_2$

or, $(AX_1)^t = (\lambda_1 X_1)^t$

or, $X_1^t A^t = \lambda_1 X_1^t$

or, $X_1^t A = \lambda_1 X_1^t$

$X_1^t A X_2 = \lambda_1 X_1^t X_2$

$X_1^t \lambda_2 X_2 = \lambda_1 X_1^t X_2$

$\lambda_2 X_1^t X_2 = \lambda_1 X_1^t X_2$

or, $(\lambda_2 - \lambda_1) X_1^t X_2 = 0$

$X_1^t X_2 = 0$ (since, $\lambda_1 \neq \lambda_2$)

$$\begin{array}{ccc} \lambda_1 & & \lambda_2 \\ \downarrow & & \downarrow \\ X_1 & & X_2 \end{array}$$

Theorem: Each eigen value of a real orthogonal matrix has unit modulus.

Ex) If λ be an eigen value of a real orthogonal matrix; prove that $1/\lambda$ is also an eigen value of A.

Soln: Let A be an orthogonal matrix of order n. Then.

$$A^T A = I_n$$

since A is non singular, $\lambda \neq 0$

$$\det(A - \lambda I_n) = 0$$

or, $\det(A - \lambda A A^T) = 0$

or, $\det A \cdot \det(I_n - \lambda A^T) = 0$

or, $\det(I_n - \lambda A^T) = 0$, since $\det A \neq 0$

or, $\lambda^n \det\left(\frac{1}{\lambda} I_n - A^T\right) = 0$

or, $(-1)^n \lambda^n \cdot \det\left(A^T - \frac{1}{\lambda} I_n\right) = 0$

or, $\det\left(A^T - \frac{1}{\lambda} I_n\right) = 0$

$\Rightarrow$ $\frac{1}{\lambda}$ is an eigen value of A^T

$\Rightarrow$ $1/\lambda$ is an eigen value of A.

Ex) A is a 3×3 real matrix having eigen values $2, 3, 1$. $\begin{pmatrix}1\\2\\1\end{pmatrix}, \begin{pmatrix}0\\1\\1\end{pmatrix}, \begin{pmatrix}1\\1\\1\end{pmatrix}$ are the eigen vectors of A corresponding to the eigen values $2, 3, 1$ respectively.

Find the matrix A.

$\underline{\text{Sol}^n}$: Let $X_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $X_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, $X_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

Then $AX_1 = \begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix}$, $AX_2 = \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix}$, $AX_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Let $P = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ Then, $AP = \begin{pmatrix} 2 & 0 & 1 \\ 4 & 3 & 1 \\ 2 & 3 & 1 \end{pmatrix}$

The column vectors of P are the eigen vectors of A corresponding to three distinct eigen values. Therefore P is non-singular.

$$P^{-1} = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 4 & 3 & 1 \\ 2 & 3 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & -1 \\ -2 & 3 & 0 \\ -2 & 1 & 2 \end{pmatrix}$$

Diagonalisation of Matrices:

Consider the set of all $n \times n$ matrices over the field F. An $n \times n$ matrix A is said to be similar to a matrix $B_{n\times n}$ if there exist a non-singular matrix P such that

$$B = P^{-1}AP$$
$$A = PBP^{-1}$$

$\underline{\text{For example}}$: Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$; $B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

These two matrices have the same characteristic polynomial and hence they have the same eigen values. But A being the matrix I_2, there is no matrix other than itself, which is similar to it. because for any non-singular 2×2 matrix P,

$$P^{-1} I_2 P = I_2$$

$\therefore$ B is not similar to A.

Defn: An $n\times n$ matrix is said to be diagonalisable if A is ~~said~~ similar to a diagonal matrix.

If A is similar to a diagonal matrix $D = diag(\lambda_1, \lambda_2, \dots \lambda_n)$ then $\lambda_1, \lambda_2, \dots \lambda_n$ are the eigen values of A.

Theorem: Let A be an $n\times n$ matrix over a field F. with eigen values $d_1, d_2, d_3 \dots d_n \in F$, d_i being not necessarily all distinct.

Let $D = diag(d_1, d_2, d_3 \dots d_n)$. then for an $n\times n$ matrix P, $AP = PD$ holds if and only if the i^{th} column vector of P be an eigen vector of A corresponding to d_i.

$$P = (X_1\ X_2\ X_3) \quad D = (\lambda_1, \lambda_2, \lambda_3) \quad (\lambda_1, X_1)(\lambda_2, X_2)(\lambda_3, X_3)$$

Proof: $AP = PD$ hold iff the i^{th} column vector of AP be equal to the i^{th} column vector of PD. for $i = 1, 2, \dots n$

Let $P = (P_{ij})_{n\times n}$. Then i^{th} column of AP and PD are

$$A\begin{pmatrix} P_{1i} \\ P_{2i} \\ \vdots \\ P_{ni} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} P_{1i} d_i \\ P_{2i} d_i \\ \vdots \\ P_{ni} d_i \end{pmatrix} \text{ respectively.}$$

$AP = PD$ iff

$$A\begin{pmatrix} P_{1i} \\ P_{2i} \\ \vdots \\ P_{ni} \end{pmatrix} = d_i \begin{pmatrix} P_{1i} \\ P_{2i} \\ \vdots \\ P_{ni} \end{pmatrix} \quad \text{for } i = 1, 2 \dots n$$

i.e., iff $\begin{pmatrix} P_{1i} \\ P_{2i} \\ \vdots \\ P_{ni} \end{pmatrix}$ is an eigen vector of A.

Theorem: A $n\times n$ matrix A over a field F is diagonalisable iff there exists n eigen vectors of A which are linearly independent.

Ex) $A = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

Then the characteristic equation of A is $(x-1)^2(x-5) = 0$.

The eigen values of A are $1, 1, 5$.

The eigen vectors corresponding to the eigen value 1 are the non-null solutions of the system of equations

$$AX = 1X$$

$$\Rightarrow \begin{aligned} 2x_1 + 2x_2 + x_3 &= 0 \\ 2x_1 + 2x_2 + x_3 &= 0 \\ x_3 &= x_3 \checkmark \end{aligned}$$

The system is equivalent to

$$x_1 + x_2 + \tfrac{1}{2}x_3 = 0$$

Take $x_2 = -c$, $x_3 = -2d$, $x_1 = -x_2 - \frac{1}{2}x_3 = c+d$

$$\begin{pmatrix} c+d \\ -c \\ -2d \end{pmatrix}$$

The eigen vectors are,

$$c\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + d\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \quad \text{where } (c,d) \neq (0,0)$$

Two linearly independent eigen vectors corresponding to the eigen value 1 are $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$

The eigen vectors corresponding to the eigen value 5 are the non-null solutions of $AX = 5X$ or,

$$(A - 5I_3)X = 0$$

$$-2x_1+2x_2+x_3=0$$
$$2x_1-2x_2+x_3=0$$
$$x_3=0$$

The system is equivalent to $x_1-x_2=0$, $x_3=0$.

The eigen vectors are $c\begin{pmatrix}1\\1\\0\end{pmatrix}$.

If $P=\begin{pmatrix}1&1&1\\-1&0&1\\0&-2&0\end{pmatrix}$, then $P^{-1}AP=\text{diag}(1,1,5)$

Ex ① Find a matrix P such that $P^{-1}AP$ is a diagonal matrix where $A=\begin{pmatrix}1&1&-2\\-1&2&1\\0&1&-1\end{pmatrix}$

Hints: Eigen values of A — $1,2,-1$ $\quad c_1\begin{pmatrix}3\\2\\1\end{pmatrix}, c_2\begin{pmatrix}1\\3\\1\end{pmatrix}, c_3\begin{pmatrix}1\\0\\1\end{pmatrix}$

$P=\begin{pmatrix}3&1&1\\2&3&0\\1&1&1\end{pmatrix}$, Then P is a non-singular matrix. $AP=PD$.

$AP=(\ )(\ )$, $PD=$

$P^{-1}AP=\text{diag}(1,2,-1)$

Ex ② Diagonalize the matrix A, where $A=\begin{pmatrix}1&-3&3\\3&-5&3\\6&-6&4\end{pmatrix}$

Soln: ① $\begin{vmatrix}1-x&1&-2\\-1&2-x&1\\0&1&-1-x\end{vmatrix}=$ $\cancel{(1-x)(x-3)+(-1)+(x+2)(1+x)}$

$\cancel{-x^2-3+4x+1}$ $\quad (x+1)(x-2)$

$-\cancel{x^2}+4\cancel{x}-2.$

~~Eigen values: $-4\pm\sqrt{16-8}$~~

$(1-x)(x^2-x-3)-(x+1)+2=-x^3+x^2+3x+x^2-x-3-x-1+2$

$=-x^3+2x^2+x-2$

1	−1	2	1	−2
	0	−1	1	2
	−1	1	2	0

$-x^2+x+2=0$

$\frac{-1\pm\sqrt{1+8}}{-2}=\frac{1\pm3}{2}$

$2,-1$

Eigen values: $1,2,-1$ ✓

Eigen vectors corresponding to 1

$$\left.\begin{array}{r} x_2 - 2x_3 = 0 \\ -x_1 + x_2 + x_3 = 0 \\ x_2 = 2x_3 \end{array}\right\} \quad \begin{array}{l} x_1 = 3x_3 \\ x_2 = 2x_3 . \end{array} \qquad c_1\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}, \ c_1 \in R, c_1 \neq 0$$

Eigen vectors corresponding to -1

$$\left.\begin{array}{r} 2x_1 + x_2 - 2x_3 = 0 \\ -x_1 + 3x_2 + x_3 = 0 \\ x_2 = 0 \end{array}\right\} \quad \begin{array}{l} x_1 = x_3 \\ x_2 = 0 \end{array} \qquad c_3\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \ c_3 \in R, c_3 \neq 0$$

Eigen vectors corresponding to 2

$$\left.\begin{array}{r} -x_1 + x_2 - 2x_3 = 0 . \\ -x_1 + x_3 = 0 \\ x_2 - 3x_3 = 0 \end{array}\right\} \quad \begin{array}{l} x_1 = x_3 \\ x_2 = 3x_3 \end{array} \qquad c_2\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \ c_2 \in R, c_2 \neq 0$$

$$\therefore P = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 3 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad P \text{ is a non-singular matrix.}$$

$$AP = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}\begin{pmatrix} 3 & 1 & 1 \\ 2 & 3 & 0 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 2 & -1 \\ 2 & 6 & 0 \\ 1 & 2 & -1 \end{pmatrix}.$$

$$PD = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 3 & 0 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 2 & -1 \\ 2 & 6 & 0 \\ 1 & 2 & -1 \end{pmatrix}$$

$$AP = PD .$$

$$P^{-1}AP = D = \operatorname{diag}(1, 2, -1).$$

---

(2) $$\begin{vmatrix} 1-x & -3 & -3 \\ 3 & -5-x & 3 \\ 6 & -6 & 4-x \end{vmatrix} = (1-x)(x^2+x-20+18) - 3(18-12+3x) + 3(-18+6x+30)$$

$$\cancel{x^2} + \cancel{x} - 2 - x^3 - \cancel{x^2} + \cancel{2x} - \cancel{18} - \cancel{9x} + \cancel{18x} + \cancel{36}$$

$(x-4)(x+2)$

-2	-1	0	12	16
	0	$+2$	-4	-16
	-1	$+2$	8	0

$$-x^3 + 12x + 16 = 0$$

$$-x^2 + 2x + 8 = 0$$

$$\frac{-2 \pm \sqrt{4+32}}{-2}$$

$$\frac{2 \pm 6}{2} = 4, -2$$

Eigen values : $-2, 4, -2$.

Eigen vectors corresponding to -2.

$$\left.\begin{aligned} 3x_1 - 3x_2 + 3x_3 &= 0 \\ 3x_1 - 3x_2 + 3x_3 &= 0 \\ 6x_1 - 6x_2 + 6x_3 &= 0 \end{aligned}\right\} \quad x_1 + x_3 = x_2$$

$$c\begin{pmatrix}1\\1\\0\end{pmatrix} + d\begin{pmatrix}0\\1\\1\end{pmatrix} \cdot \quad c, d \in \mathbb{R} ; \ c, d \neq 0.$$

Eigen vectors corresponding to 4

$$\left.\begin{aligned} -3x_1 - 3x_2 + 3x_3 &= 0 \\ 3x_1 - 9x_2 + 3x_3 &= 0 \\ 6x_1 - 6x_2 &= 0 \end{aligned}\right\} \quad \begin{aligned} x_1 &= x_2 \\ x_3 &= 2x_2 \end{aligned} \qquad c_1\begin{pmatrix}1\\1\\2\end{pmatrix}, \ c_1 \in \mathbb{R}, c_1 \neq 0$$

$$\therefore P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}. \quad P \text{ is a non-singular matrix}.$$

$$AP = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} -2 & 0 & 4 \\ -2 & -2 & 4 \\ 0 & -2 & 8 \end{pmatrix}$$

$$PD = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}\begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} -2 & 0 & 4 \\ -2 & -2 & 4 \\ 0 & -2 & 8 \end{pmatrix}$$

$$P^{-1}AP = D = \text{diag}(-2, -2, 4)$$

Orthogonal diagonalisation of real matrices:

A square matrix A is said to be orthogonally diagonalisable if $\exists$ an orthogonal matrix P such that $P^{-1}AP$ is a diagonal matrix. The matrix P is said to diagonalise A orthogonally.

Theorem: A square matrix A is orthogonally diagonalisable iff A is symmetric.

Proof: Let A be orthogonally diagonalisable. Then $\exists$ an orthogonal matrix P such that $P^{-1}AP$ is diagonal.

Let $P^{-1}AP = D$ where D is a diagonal matrix.

$$\text{Then } A = PDP^{-1} = PDP^T$$

$$\begin{aligned} A^T &= (PDP^T)^T \\ &= (P^T)^T D^T P^T \\ &= PDP^T \\ &= A \end{aligned}$$

Ex) Diagonalise the matrix $A = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}$ orthogonally.

Soln: The eigen values of A are $1, -2, 4$.

The eigen vectors corresponding to the eigen values $1, -2, 4$ are respectively $c_1\begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$, $c_2\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and $c_3\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$ where c_1, c_2, c_3 are arbitrary non-zero real numbers.

Taking one eigen vector corresponding to each eigen value, we get the orthogonal eigen vectors.

$$\alpha = \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}, \beta = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \gamma = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

The corresponding orthonormal eigen vectors are $\frac{1}{3}\alpha$, $\frac{1}{3}\beta$ and $\frac{1}{3}\gamma$.

$$\text{Let } P = \begin{pmatrix} -2/3 & 1/3 & 2/3 \\ -1/3 & 2/3 & -2/3 \\ 2/3 & 2/3 & 1/3 \end{pmatrix}$$

$$AP = \frac{1}{3}\begin{pmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{pmatrix}\begin{pmatrix} -2 & 1 & 2 \\ -1 & 2 & -2 \\ 2 & 2 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -2 & -2 & 8 \\ -1 & -4 & -8 \\ 2 & -4 & 4 \end{pmatrix}$$

$$= \frac{1}{3}\begin{pmatrix} -2 & 1 & 2 \\ -1 & 2 & -2 \\ 2 & 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{pmatrix} = PD$$

$$P^{-1}AP = \mathrm{diag}(1, -2, 4)$$

Real quadratic form:

An expression of the form $\sum_{j=1}^{n}\sum_{i=1}^{n} a_{ij}' x_i x_j$; where a_{ij} are real and $a_{ij} = a_{ji}$, is said to be a real quadratic form in n variables $x_1, x_2, \ldots x_n$.

The matrix notation for the quadratic form is

$$X^T A X,$$

where $X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ and $A = (a_{ij})_{n \times n}$.

A is a real symmetric matrix since $a_{ij} \in R$ and $a_{ij} = a_{ji}$

$$\sum_{j=1}^{n}\sum_{i=1}^{n} a_{ij}\, x_i x_j = X^T A X \quad ; \quad X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad ; \quad A = (a_{ij})_{n\times n}$$

Ex-1) $5x_1^2 + 2x_1x_2 - x_2^2$ is a real quadratic form in two variables x_1, x_2. The associated matrix is $A = \begin{pmatrix} 5 & 1 \\ 1 & -1 \end{pmatrix}$

$$X^TAX = (x_1 \;\; x_2)\begin{pmatrix} 5 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (5x_1+x_2 \;\; x_1-x_2)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$= 5x_1^2 + 2x_1x_2 - x_2^2 .$$

Ex-2) $x_1x_2 - x_2x_3$

$$A = \begin{pmatrix} 0 & 1/2 & 0 \\ 1/2 & 0 & -1/2 \\ 0 & 1/2 & 0 \end{pmatrix}$$

Ex-3) $x_1^2 - x_2^2 + 2x_3^2$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Definition: A real quadratic form $Q(x_1, x_2, \ldots x_n)$ assumes the value 0 when $X = 0$. But Q takes up different real values for different non-zero X.

A real quadratic form $Q = X^TAX$ is said to be

i) positive definite if $Q > 0$, for all $X \neq 0$

ii) positive semi-definite if $Q \geq 0$, for all X and $Q = 0$ for some $X \neq 0$

iii) negative definite if $Q < 0$ for all $X \neq 0$

iv) negative semi-definite if $Q \leq 0$ for all X and $Q = 0$ for some $X \neq 0$

v) indefinite if $Q \geq 0$ for some $X \neq 0$ and $Q \leq 0$ for some other $X \neq 0$.

Ex-4) Let $Q(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 4x_3^2 + 2x_1x_2 - 4x_2x_3 - 2x_3x_1$

$$= (x_1 + x_2 - x_3)^2 + (x_2 - x_3)^2 + 2x_3^2$$

$x_1^2 + (x_2 - x_3)^2 + 2x_1x_2 - 2x_3x_1$

$Q \geq 0$ for all (x_1, x_2, x_3) and

$Q = 0$ only when $x_1 = x_2 = x_3 = 0$

Q is positive definite.

The associated matrix $\begin{pmatrix} 1 & 1 & -1 \\ 1 & 2 & -2 \\ -1 & -2 & 4 \end{pmatrix}$.

Ex-5) Let $Q(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 - 4x_2x_3 - 2x_3x_1$

$$= (x_1 + x_2 - x_3)^2 + (x_2 - x_3)^2$$

$Q \geq 0 \ \forall \ (x_1, x_2, x_3)$

But Q may be zero for non-zero (x_1, x_2, x_3)

For example, $Q(0,1,1) = 0$.

$\therefore$ Q is positive semi-definite.

Ex-6) Let $Q(x_1, x_2, x_3) = x_1^2 + x_2^2 + 2x_1x_2 + 2x_2x_3 + 2x_3x_1 + x_3^2 - x_3^2$

$$= (x_1 + x_2 + x_3)^2 - x_3^2$$

Q is indefinite because

$Q(1,1,0) > 0$, $Q(0,-1,1) < 0$.

$Q = X^T A X$

$X = PY$

$Q = (PY)^T A (PY)$

$= Y^T P^T A P Y = Y^T B Y$

This is a quadratic form in Y, since $P^T A P$ is a real symmetric matrix.

$(P^TAP)^T = P^TA^T(P^T)^T = P^TAP$.

Thus Q is transformed to a quadratic form Q' with associated matrix.

For a real symmetric matrix A of rank $r (\leq n)$, there exists a non-singular matrix P, such that P^TAP becomes a diagonal matrix

$$D = \begin{pmatrix} I_m & & \\ & -I_{r-m} & \\ & & 0 \end{pmatrix}$$ of rank r, where $0 \leq m \leq r$.

By a suitable transformation $X = PY$, where P is non-singular, the real quadratic form Q transforms to

$$y_1^2 + y_2^2 + \cdots + y_m^2 - y_{m+1}^2 - y_{m+2}^2 - \cdots - y_r^2,$$

This is said to be the $\underbrace{\text{normal form}}$ of Q.

↳ where $0 \leq m \leq r \leq n$.

iv) If $\lambda < 0$, $m = 0$, then

$D = -y_1^2 - y_2^2 - \cdots - y_n^2$, $r < n$.

Application to the solution of certain systems of differential equations

Let $X(t)$ be a column matrix $X(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$

Consider the system of differential equations $- \; X' = AX$

$$\begin{pmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{pmatrix} = A \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

$$\frac{d(x_1(t))}{dt} = x_1 + 2x_2$$

$$\frac{d(x_2(t))}{dt} = 3x_1 + 7x_2$$

$$\begin{pmatrix} dx_1/dt \\ dx_2/dt \end{pmatrix} = \begin{pmatrix} \; \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$

We want to find a solution such that $X(0) = X_0$ (a given column matrix).

Make coordinate transformation. $Y = PX$

$Y' = PX'$

$$X' = P^{-1}Y' = AP^{-1}Y$$

or, $Y' = (PAP^{-1})Y$

$\exists$ a P such that $PAP^{-1} = D$, a diagonal matrix.

Hence, the new system is $\frac{dy_1}{dt} = \lambda_1 y_1$, $\frac{dy_2}{dt} = \lambda_2 y_2$ ---- $\frac{dy_n}{dt} = \lambda_n y_n$ $\rightarrow$ (2)

where $\lambda_1, \lambda_2, --- \lambda_n$ are characteristic values of A.

Solution of (2) is

$$y_1 = c_1 e^{\lambda_1 t}, \; y_2 = c_2 e^{\lambda_2 t} \; --- \; y_n = c_n e^{\lambda_n t}$$

$$X(t) = P^{-1}\begin{pmatrix} c_1 e^{\lambda_1 t} \\ c_2 e^{\lambda_2 t} \\ \vdots \\ c_n e^{\lambda_n t} \end{pmatrix} = c_1 e^{\lambda_1 t} X_1 + c_2 e^{\lambda_2 t} X_2 + \cdots c_n e^{\lambda_n t} X_n$$

At $t = 0$, we have

$$X_0 = X(0) = P^{-1}\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = c_1 X_1 + c_2 X_2 + --- c_n X_n$$

where, $X_1, X_2, ----X_n$ are characteristic vectors of A.

Ex) Find a solution of the following system of differential equations.

$$\frac{dx_1}{dt} = 2x_1 + 3x_2 + 3x_3 \qquad \frac{dx_2}{dt} = 3x_1 - x_2 \qquad \frac{dx_3}{dt} = 3x_1 - x_3$$

Satisfying the initial conditions $x_1(0) = 1$, $x_2(0) = -2$, $x_3(0) = 0$

Soln: The system can be written as $X' = AX$, where

$$A = \begin{pmatrix} 2 & 3 & 3 \\ 3 & -1 & 0 \\ 3 & 0 & -1 \end{pmatrix}$$ is a real symmetric matrix.

The characteristic equation is

$$\begin{vmatrix} 2-\lambda & 3 & 3 \\ 3 & -1-\lambda & 0 \\ 3 & 0 & -1-\lambda \end{vmatrix} = 0$$

or, $(\lambda+1)(\lambda+4)(-\lambda+5)=0$

The characteristic values are $\lambda_1=-1, \lambda_2=-4, \lambda_3=+5$.

The corresponding eigen vectors are..

$$X_1=\begin{pmatrix}0\\1\\-1\end{pmatrix}, X_2=\begin{pmatrix}-1\\1\\1\end{pmatrix}, X_3=\begin{pmatrix}2\\1\\1\end{pmatrix}$$

A solution is,

$$x_1(t)=-C_2e^{-4t}+2C_3e^{5t}$$

$$x_2(t)=C_1e^{-t}+C_2e^{-4t}+C_3e^{5t}$$

$$x_3(t)=-C_1e^{-t}+C_2e^{-4t}+C_3e^{5t}$$

$$X(0)=(1,-2,0)$$

$$\begin{pmatrix}1\\-2\\0\end{pmatrix}=\begin{pmatrix}0&-1&2\\1&1&1\\-1&1&1\end{pmatrix}\begin{pmatrix}C_1\\C_2\\C_3\end{pmatrix}$$

$$\Rightarrow C_1=-1, C_2=-1, C_3=0$$

The solution satisfying the initial condition is

$$x_1(t)=e^{-4t}$$

$$x_2(t)=-e^{-t}-e^{-4t}$$

$$x_3(t)=e^{-t}-e^{-4t}$$

Complex Matrices:

$$A=P+iQ$$

$$\bar{A}=P-iQ$$

$$(\bar{A})^t=A^\circ$$

Hermitian matrix and Skew Hermitian Matrix:

A complex $n\times n$ matrix A is said to be Hermitian if $A^\circ=A$, and skew Hermitian if $A^\circ=-A$

Hermitian Matrix:

$$\begin{pmatrix} 1 & 2+i & -1 \\ 2-i & 2 & i \\ -1 & -i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3i \\ 2 & 0 & 4 \\ -3i & 4 & 2 \end{pmatrix}$$

Skew-Hermitian Matrix:

$$\begin{pmatrix} 0 & 2 & i \\ -2 & 0 & 1+i \\ i & -1+i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2i & 1+2i \\ 2i & i & 1 \\ -1+2i & -1 & 0 \end{pmatrix}$$

Theorem:

If $H = P + iQ$ be a Hermitian matrix, then

i) The diagonal elements of H are all real numbers

ii) P is a real symmetric matrix and Q is a real skew-symmetric matrix.

Theorem:

If $S = M + iN$

Skew-Hermitian matrix.

i) The diagonal elements of S are zero or purely imaginary

ii) M is a real skew symmetric matrix and N is a real symmetric matrix.

Tutorial: -2

1) Determine the nature, index and signature of $x^2+2y^2+2z^2+2x-2yz-2xy$

2) Obtain the symmetric matrix for the quadratic form

$$x_1^2+2x_1x_2+6x_2x_3-4x_1x_3-5x_2^2+4x_3^2$$

3) Solve the following differential equations (initial value problem) by matrix method.

$$x''-2x'-3x=0$$

$$x(0)=0,\ x'(0)=1$$